

① A coin is tossed 10 times and the outcomes are:

HHT HHT HHT

Using Maximum likelihood Estimation, estimate the probability of getting Heads (P).

① Given:

Outcomes: HHT HHT HHT

Total tosses (n) = 10

Number of Heads (x) = 7

Formula:

$$\hat{p} = \frac{x}{n}$$

Calculation:

$$\hat{p} = \frac{7}{10} = 0.7$$

∴ Estimated probability of getting Heads = 0.7 (70%)

② Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

① Given:

Total students (n) = 50

students passed (x) = 42

Formula:

$$\hat{p} = \frac{x}{n}$$

calculation:

$$\hat{p} = \frac{42}{50} = 0.84$$

$\therefore$ MLE of passing probability = 0.84 (84%)

- ③ A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.

Find the MLE for the probability that a bulb is defective.

④ Given,

Total bulbs (n) = 100

Defective bulbs (x) = 8

Formula:

$$\hat{p} = \frac{x}{n}$$

Calculation:

$$\hat{p} = \frac{8}{100} = 0.08$$

$\therefore$ Probability that a bulb is defective = 0.08 (8%)

- ④ A biased die is rolled 60 times. The number 6 appears 18 times.

Estimate the probability of rolling a 6 using MLE.

④ Given,

$$\text{Total rolls } (n) = 60$$

$$\text{Number of times 6 appears } (x) = 18$$

Formula:

$$\hat{p} = \frac{x}{n}$$

Calculation:

$$\hat{p} = \frac{18}{60} = 0.30$$

$\therefore$ Estimated probability of rolling a 6 = 0.30 (30%)

⑤ Given:

$$\text{Input } (x) = 4$$

$$\text{weight } (w) = 2$$

$$\text{Actual output} = 10$$

prediction:

$$y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, Find the updated weight.

(A)

Given:

$$\text{input } (x) = 4$$

$$\text{weight } (w) = 2$$

$$\text{Actual output} = 10$$

prediction formula:

$$y = w \times x$$

$$\text{Gradient} = 4$$

$$\text{Learning Rate} = 0.01$$

① Find the predicted output

Formula:

$$y = w \times x$$

Calculation:

$$y = 2 \times 4 = 8$$

$$\therefore \text{predicted output} = 8$$

② Calculate the Error

Formula:

$$\text{Error} = \text{Actual output} - \text{Predicted output}$$

Calculation:

$$\Rightarrow 10 - 8$$

$$\Rightarrow 2$$

$$\therefore \text{Error} = 2$$

③ Update the Weight:-

Formula:

$$W_{\text{new}} = W - (\text{Learning Rate} \times \text{Gradient})$$

Substitute the values:

$$W_{\text{new}} = 2 - (0.01 \times 4)$$

Multiply:

$$0.01 \times 4 = 0.04$$

Final Calculation:

$$W_{\text{new}} = 2 - 0.04 = 1.96$$

$$\therefore \text{Updated weight} = 1.96$$